

Lab: Your Brain as Agent -- The Command Center

Objective

Learning Goal: Investigate how your brain acts as an agent -- receiving sensory information, building models of the body and world, and sending commands to keep you alive and functioning.

This is an **investigation lab** designed for middle school students (Grades 6-8).

Materials Needed

- Ruler (for reaction time test)
 - Blindfold or eye mask
 - Paper and colored pencils
 - Lab journal or notebook
-

Part 1: Reaction Time Test (10 min)

Your brain is an agent that constantly predicts what is about to happen and prepares actions in advance. Let's measure how fast your brain-agent reacts.

Ruler Drop Test: Partner A holds a ruler vertically. Partner B places their thumb and finger at the 0 cm mark (without touching). Partner A drops the ruler without warning. Partner B catches it. Record the centimeter mark where they caught it.

Trial	Catch Position (cm)	Were you expecting the drop?	Prediction helped?
1		No (surprise)	
2		Somewhat	
3		Getting better	
4			
5			
Average			

Write your response here

Part 2: Brain Agent Profile (15 min)

Your brain is the ultimate agent. It has sensors (your senses), an internal model (your mental picture of the world), and actions (motor commands). Map your brain as an agent.

Agent Feature	Your Brain's Version	Example
Sensors	Eyes, ears, skin, nose, tongue + internal sensors	
Internal model	Your mental picture of your body and the world	
Goals	Keep you alive, safe, comfortable, and happy	
Predictions	What you expect to feel, see, hear next	
Actions	Muscle commands, hormone releases, reflexes	
Surprise handlers	Startle reflex, pain response, attention shift	

Write your response here

Part 3: Reflex vs. Decision (10 min)

Some of your brain's actions are automatic (reflexes) and some require conscious thought (decisions). Sort these actions:

Action	Reflex or Decision?	Which part of the brain handles it?	How fast?
Pulling your hand off a hot stove			
Choosing what to eat for lunch			
Your pupils shrinking in bright light			
Solving a math problem			
Catching a ball thrown at your face			
Deciding whether to raise your hand in class			

Write your response here

Part 4: Sensory Deprivation Challenge (5 min)

Close your eyes for 60 seconds in a quiet space. Pay attention to what your brain does when it loses visual input.

Observation	What happened?
What did you "see" with your eyes closed?	
Did your other senses get stronger?	
What did your brain predict would happen?	
How did you feel (calm, anxious, bored)?	

Write your response here

Reflection Table

Question	Your Answer
In what ways is your brain an agent?	
What is the difference between a reflex and a decision?	
How does prediction make your reactions faster?	
What happens when your brain-agent is surprised?	
How does Active Inference describe the brain?	

Write your response here